

Inference for logistic regression models

Agenda

- + For homework submissions:
 - + Make sure to submit rendered HTML
 - + If writing math in Quarto (e.g. regression models, maximum likelihood problems, etc.), make sure the equations display correctly
- + Discuss feedback survey
- + Recap activity from end of last class
- + More inference!

Last time: Activity

$$Admit_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

We want to test whether there is a relationship between Rank and grad school acceptance, after accounting for GPA and GRE. What are my null and alternative hypotheses?

Activity

$$Admit_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

What is the reduced model?

Activity

```
##  
## Call: glm(formula = admit ~ gpa + gre + factor(rank), family = binomial,  
##      data = grad_app)  
##  
## Coefficients:  
##      (Intercept)              gpa              gre factor(rank)2 factor(rank)3  
##      -3.989979         0.804038         0.002264        -0.675443        -1.340204  
## factor(rank)4  
##      -1.551464  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  394 Residual  
## Null Deviance:          500  
## Residual Deviance: 458.5      AIC: 470.5
```

```
##  
## Call: glm(formula = admit ~ gpa + gre, family = binomial, data = grad_app)  
##  
## Coefficients:  
##      (Intercept)              gpa              gre  
##      -4.949378         0.754687         0.002691  
##  
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual  
## Null Deviance:          500  
## Residual Deviance: 480.3      AIC: 486.3
```

Activity

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Drop-in-deviance test: $G = 21.8$

What is the distribution of G under the null hypothesis?

Activity

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Drop-in-deviance test: $G = 21.8$

Under H_0 , $G \sim \chi_3^2$

```
pchisq(21.8, df=3, lower.tail=F)
```

```
## [1] 7.178958e-05
```

A new question

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GRE_i$$

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.901344	0.606038	-4.787	1.69e-06
gre	0.003582	0.000986	3.633	0.00028

Here $\hat{\beta}_1 = 0.003582$. Will $\hat{\beta}_1$ be exactly equal to the true value β_1 ?

A new question

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GRE_i$$

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Here $\hat{\beta}_1 = 0.003582$. Will $\hat{\beta}_1$ be exactly equal to the true value β_1 ?

No -- our estimate comes from one sample of data, and there is variability from sample to sample.

How can I account for this variability when reporting an estimate of β_1 ?

Interval estimates

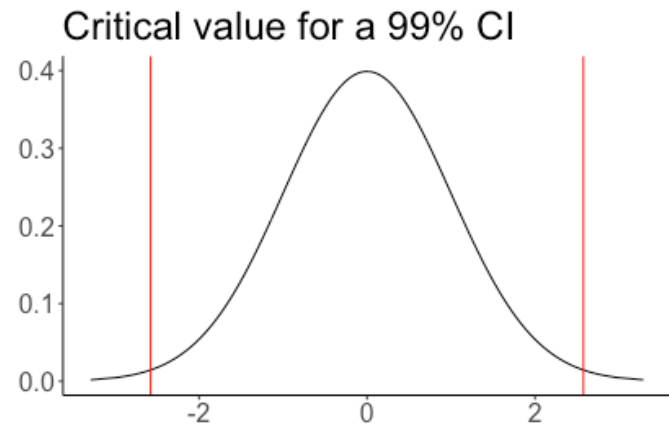
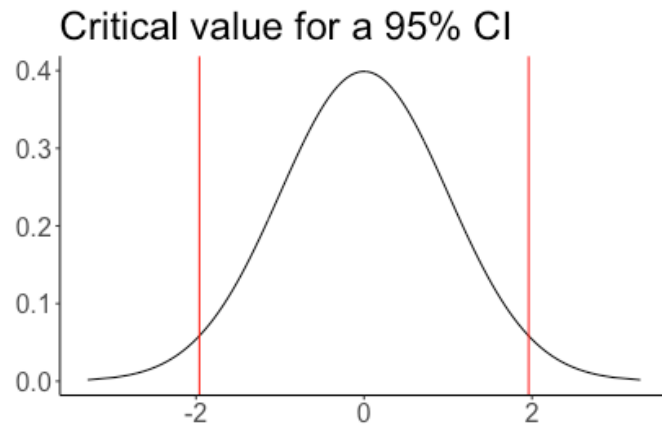
Point estimate: a single "best guess" value of a parameter. The estimated slope $\hat{\beta}_1$ is a point estimate of β_1

Interval estimate: a "reasonable range" of possible values for a parameter. A *confidence interval* is an interval estimate

How would I construct a confidence interval for β_1 ?

The critical value z^*

To construct a $(1 - \alpha) \cdot 100\%$ confidence interval, z^* is the value such that $P(Z > z^*) = \alpha/2$



Computing z^*

Example: for a 95% confidence interval, $z^* = 1.96$

```
qnorm(0.025, lower.tail=FALSE)
```

```
## [1] 1.959964
```

Example: for a 99% confidence interval, $z^* = 2.58$

```
qnorm(0.005, lower.tail=FALSE)
```

```
## [1] 2.575829
```

Confidence interval

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.901344	0.606038	-4.787	1.69e-06
gre	0.003582	0.000986	3.633	0.00028

95% confidence interval for β_1 :

Motivating the confidence interval

- + $\hat{\beta}_1$ is our "best guess" of β_1
- + But, there is variability in our estimate of β_1 ; $\hat{\beta}_1$ will vary from sample to sample

Idea: What if we take a "new" sample of data, and see how $\hat{\beta}_1$ changes?

Bootstrap sample

A **bootstrap sample** is a sample of rows from the data, *with* replacement

Original data:

GRE	Admit
380	0
660	1
800	1
520	0
760	1
640	1

Bootstrap sample:

GRE	Admit
380	0
380	0
800	1
520	0
760	1
760	1

Motivation: Variability in resampling from the *observed* data should reflect variability in resampling from the *population*

Bootstrap sample

A **bootstrap sample** is a sample of rows from the data, *with* replacement

Original data:

GRE	Admit
380	0
660	1
800	1
520	0
760	1
640	1

Another bootstrap sample:

GRE	Admit
380	0
760	1
520	0
800	1
520	0
800	1

Motivation: Variability in resampling from the *observed* data should reflect variability in resampling from the *population*

Activity

- + Take many bootstrap samples
- + For each bootstrap sample, calculate $\hat{\beta}_1$
- + Examine the distribution of estimated slopes

https://sta214-f25.github.io/class_activities/ca_15.html

Activity

```
nsim <- 1000
boot_ests <- rep(NA, nsim)

for(i in 1:nsim){
  grad_boot <- grad_app |>
    slice_sample(n = nrow(grad_app), replace=T)

  m_boot <- glm(admit ~ gre, family = binomial,
                data = grad_boot)
  boot_ests[i] <- coef(m_boot)[2]
}
```

Activity

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.901344	0.606038	-4.787	1.69e-06
gre	0.003582	0.000986	3.633	0.00028

```
mean(boot_ests)
```

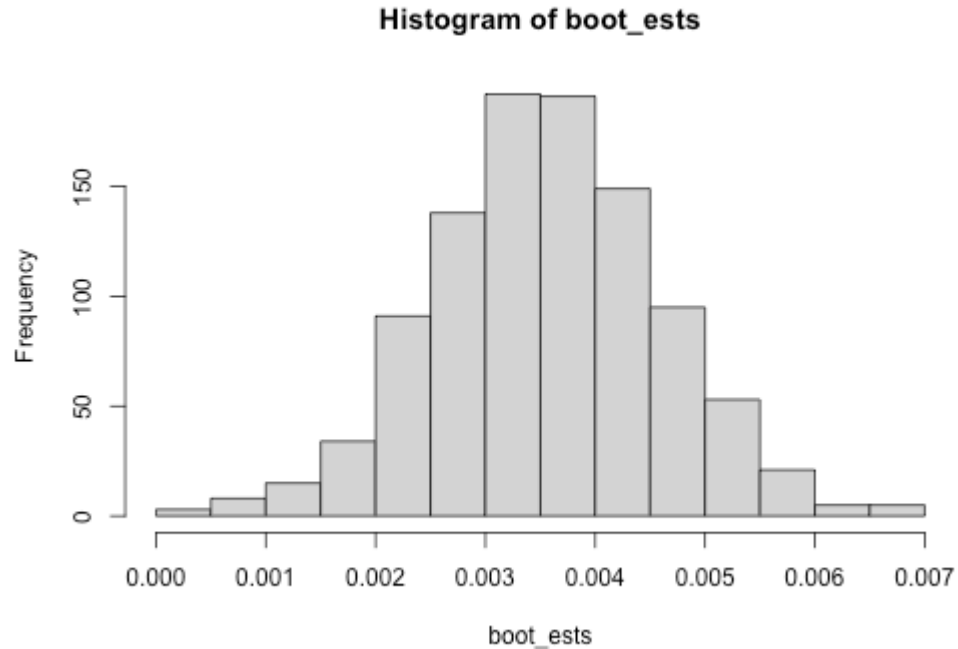
```
## [1] 0.003555198
```

```
sd(boot_ests)
```

```
## [1] 0.001040903
```

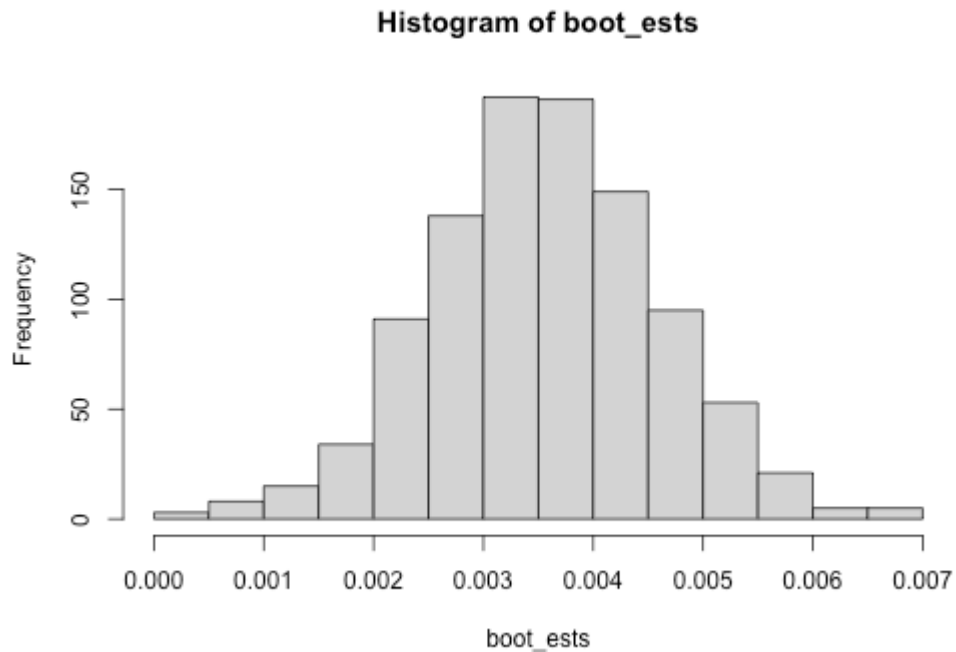
How do the mean and standard deviation of the bootstrap estimates compare to the regression output?

Activity



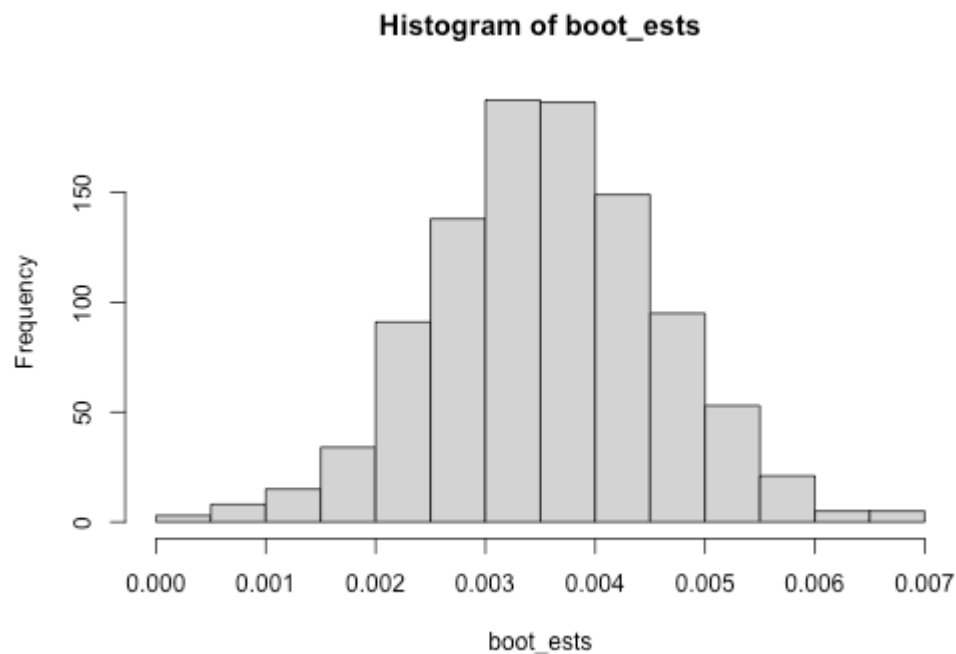
What does the bootstrap distribution look like?

Activity



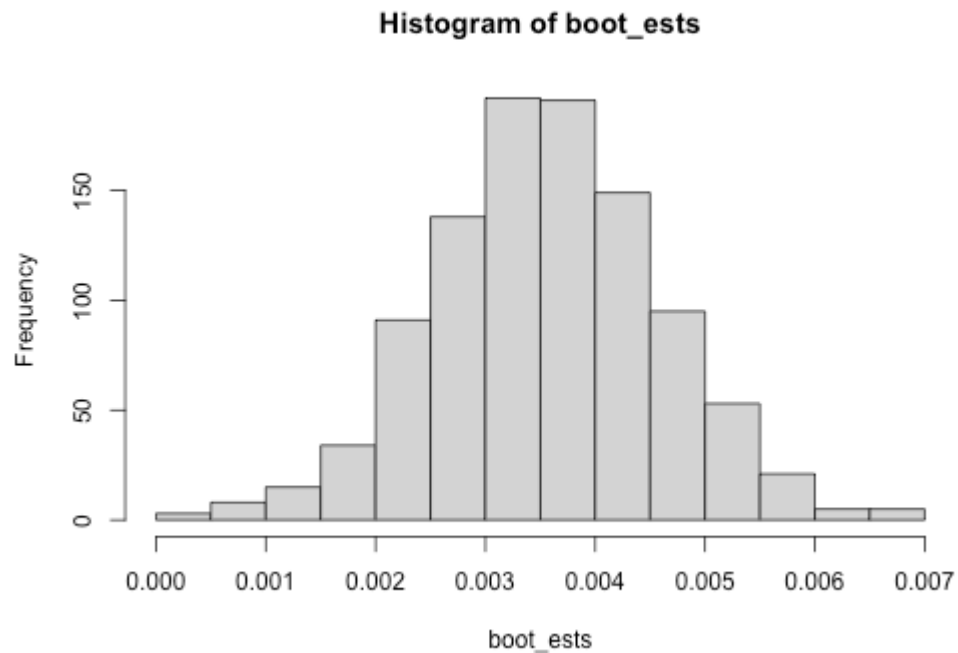
Based on the bootstrap distribution, is a value of 0.004 a "reasonable" guess for β_1 ?

Activity



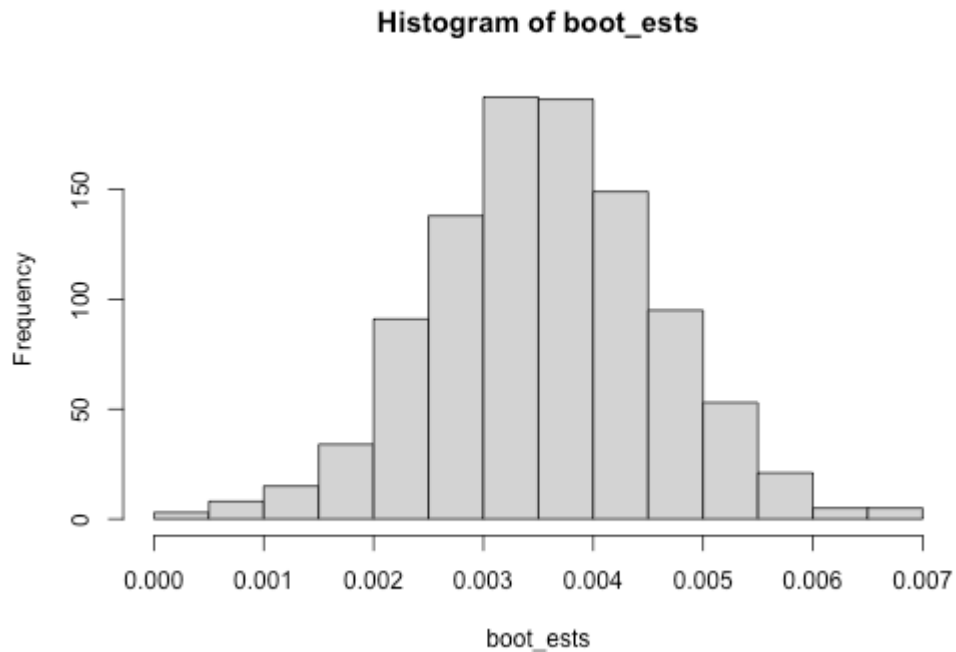
Based on the bootstrap distribution, is a value of 0.0025 a "reasonable" guess for β_1 ?

Activity



Based on the bootstrap distribution, is a value of 0.1 a "reasonable" guess for β_1 ?

Activity

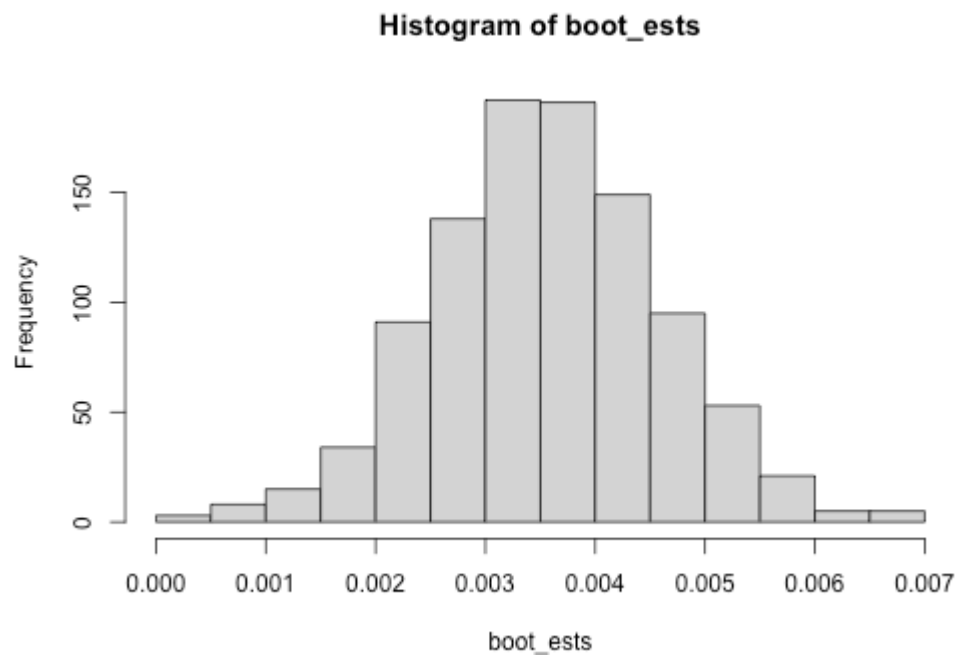


Based on the bootstrap distribution, what would be a "reasonable range" of values for β_1 ?

Confidence interval

95% confidence interval for β_1 :

$$\hat{\beta}_1 \pm 1.96SE(\hat{\beta}_1) = 0.00358 \pm 1.96(0.00099) = (0.0016, 0.0055)$$



Activity

Work on the activity (handout).

Activity

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.73743	0.08499	20.44	<2e-16	***
WBC	-0.36085	0.01243	-29.03	<2e-16	***

Calculate a 99% confidence interval for the change in the log odds of dengue associated with a one-unit increase in WBC.

Activity

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.73743	0.08499	20.44	<2e-16	***
WBC	-0.36085	0.01243	-29.03	<2e-16	***

How could we calculate a 99% confidence interval for the change in the *odds* of dengue associated with a one-unit increase in WBC?